Memebers : Talha Faheem and Daniyal Hikmat

BSSE(B)   Semester 4

Subject: DATABASE (Lab)

Multi-Vendor E-Commerce Platform with Customer and Seller Management System

Milestone	Version	Date	Remarks
ERD & DB Schema	1.0	26-04-2026	
Update Schema	1.1	30-04-2026	
Normalization	2.0	12-05-2026	
Data flow chart	3.0	12-05-2026	
Dataset Preprocessing	4.0	15-05-2026	
Database Setup (DDL)	5.0	15-05-2026	
Data Population (DML)	6.0	17-05-2026	

Git hub link: https://github.com/Talha-Faheem/e-commerce-

Database Normalization Document

ROLES Table

Attribute	Data Type	Key
id	INT	PK
role_name	VARCHAR(50)	

USER_STATUS Table

Attribute	Data Type	Key
id	INT	PK
status_name	VARCHAR(50)	

USERS Table

Attribute	Data Type	Key
id	INT	PK
name	VARCHAR(100)	
email	VARCHAR(150)	UNIQUE
password	VARCHAR(255)	
phone	VARCHAR(20)	
profile_image	VARCHAR(255)	
role_id	INT	FK
status_id	INT	FK
last_login	TIMESTAMP	
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

SELLERS Table

Attribute	Data Type	Key

id	INT	PK
user_id	INT	FK
shop_name	VARCHAR(150)	
shop_logo	VARCHAR(255)	
shop_banner	VARCHAR(255)	
shop_description	TEXT	
cnic	VARCHAR(30)	
is_verified	BOOLEAN	
rating	DECIMAL	
total_sales	INT	

SELLER_BANKS Table

Attribute	Data Type	Key
id	INT	PK
seller_id	INT	FK
bank_name	VARCHAR(100)	
account_number	VARCHAR(100)	
iban	VARCHAR(100)	
account_title	VARCHAR(100)	

CUSTOMERS Table

Attribute	Data Type	Key
id	INT	PK
user_id	INT	FK
loyalty_points	INT	
total_orders	INT	
wishlist_count	INT	

CATEGORIES Table

Attribute	Data Type	Key
id	INT	PK
name	VARCHAR(100)	
slug	VARCHAR(150)	UNIQUE
image	VARCHAR(255)	
description	TEXT	
status	VARCHAR(50)	

PRODUCTS Table

Attribute	Data Type	Key
id	INT	PK
seller_id	INT	FK
category_id	INT	FK
name	VARCHAR(150)	
slug	VARCHAR(200)	UNIQUE
description	TEXT	
price	DECIMAL	

discount_price	DECIMAL	
thumbnail	VARCHAR(255)	
status	VARCHAR(50)	

PRODUCT_VARIANTS Table

Attribute	Data Type	Key
id	INT	PK
product_id	INT	FK
size	VARCHAR(50)	
color	VARCHAR(50)	
sku	VARCHAR(100)	UNIQUE
price	DECIMAL	
stock	INT	

INVENTORY Table

Attribute	Data Type	Key
id	INT	PK
product_id	INT	FK
stock	INT	
reserved_stock	INT	
warehouse_location	VARCHAR(255)	
updated_at	TIMESTAMP	

PRODUCT_IMAGES Table

Attribute	Data Type	Key
id	INT	PK
product_id	INT	FK
image_url	VARCHAR(255)	

ADDRESSES Table

Attribute	Data Type	Key
id	INT	PK
user_id	INT	FK
full_name	VARCHAR(100)	
phone	VARCHAR(20)	
country_id	INT	FK
province_id	INT	FK
city_id	INT	FK
postal_code	VARCHAR(20)	
address_line_1	VARCHAR(255)	
address_line_2	VARCHAR(255)	
landmark	VARCHAR(255)	
address_type	VARCHAR(50)	

CARTS Table

Attribute	Data Type	Key

id	INT	PK
customer_id	INT	FK
created_at	TIMESTAMP	

CART_ITEMS Table

Attribute	Data Type	Key
id	INT	PK
cart_id	INT	FK
product_id	INT	FK
quantity	INT	

WISHLISTS Table

Attribute	Data Type	Key
id	INT	PK
customer_id	INT	FK

WISHLIST_ITEMS Table

Attribute	Data Type	Key
id	INT	PK
wishlist_id	INT	FK
product_id	INT	FK

ORDERS Table

Attribute	Data Type	Key
id	INT	PK
customer_id	INT	FK
address_id	INT	FK
order_number	VARCHAR(100)	UNIQUE
subtotal	DECIMAL	
shipping_fee	DECIMAL	
tax	DECIMAL	
discount	DECIMAL	
total_amount	DECIMAL	
payment_method_id	INT	FK
payment_status_id	INT	FK
order_status_id	INT	FK

ORDER_ITEMS Table

Attribute	Data Type	Key
id	INT	PK
order_id	INT	FK
product_id	INT	FK
seller_id	INT	FK
quantity	INT	
price	DECIMAL	
subtotal	DECIMAL	

PAYMENTS Table

Attribute	Data Type	Key
id	INT	PK
order_id	INT	FK
transaction_id	VARCHAR(150)	UNIQUE
payment_method_id	INT	FK
amount	DECIMAL	
payment_status_id	INT	FK

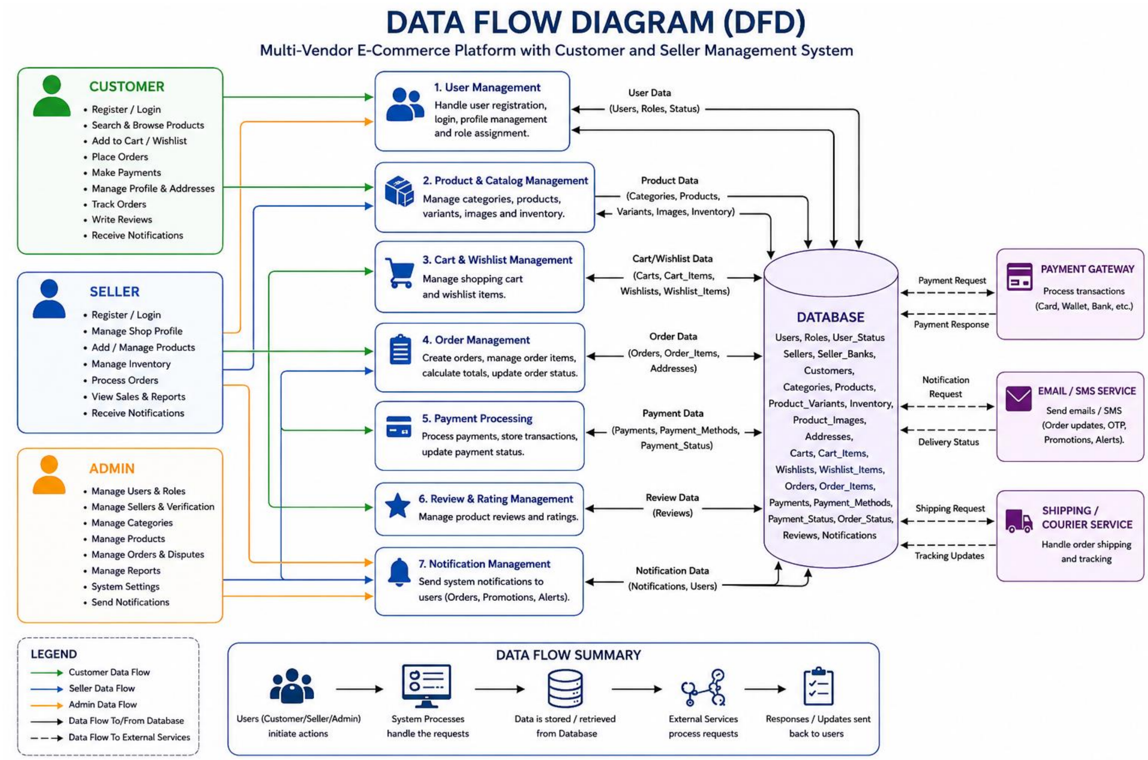
REVIEWS Table

Attribute	Data Type	Key
id	INT	PK
customer_id	INT	FK
product_id	INT	FK
rating	INT	
comment	TEXT	
UNIQUE(customer_id, product_id)		

NOTIFICATIONS Table

Attribute	Data Type	Key
id	INT	PK
user_id	INT	FK
title	VARCHAR(255)	
message	TEXT	
type	VARCHAR(50)	
is_read	BOOLEAN	

Dataset Preprocessing



Csv files on github

Database Setup (DDL)

```
CREATE DATABASE ecommerce_platform;

USE ecommerce_platform;


CREATE TABLE roles (

   id INT PRIMARY KEY AUTO_INCREMENT,

   role_name VARCHAR(50) NOT NULL UNIQUE

);


CREATE TABLE user_status (

   id INT PRIMARY KEY AUTO_INCREMENT,
```

```
    status_name VARCHAR(50) NOT NULL UNIQUE
);

CREATE TABLE users (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(150) NOT NULL UNIQUE,
    password VARCHAR(255) NOT NULL,
    phone VARCHAR(20) NOT NULL,
    profile_image VARCHAR(255),
    role_id INT NOT NULL,
    status_id INT NOT NULL,
    last_login TIMESTAMP NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,

    CONSTRAINT fk_users_role
        FOREIGN KEY (role_id) REFERENCES roles(id),

    CONSTRAINT fk_users_status
        FOREIGN KEY (status_id) REFERENCES user_status(id)
```

```
);

CREATE TABLE sellers (
    id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT NOT NULL UNIQUE,
    shop_name VARCHAR(150) NOT NULL,
    shop_logo VARCHAR(255),
    shop_banner VARCHAR(255),
    shop_description TEXT,
    cnic VARCHAR(30) NOT NULL UNIQUE,
    is_verified BOOLEAN DEFAULT FALSE,
    rating DECIMAL(3,2) DEFAULT 0,
    total_sales INT DEFAULT 0,

    CONSTRAINT chk_seller_rating
        CHECK (rating >= 0 AND rating <= 5),

    CONSTRAINT fk_sellers_user
        FOREIGN KEY (user_id) REFERENCES users(id)
);

CREATE TABLE seller_banks (
```

```sql
    id INT PRIMARY KEY AUTO_INCREMENT,
    seller_id INT NOT NULL,
    bank_name VARCHAR(100) NOT NULL,
    account_number VARCHAR(100) NOT NULL UNIQUE,
    iban VARCHAR(100) UNIQUE,
    account_title VARCHAR(100) NOT NULL,

    CONSTRAINT fk_seller_banks_seller
        FOREIGN KEY (seller_id) REFERENCES sellers(id)
);

CREATE TABLE customers (
    id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT NOT NULL UNIQUE,
    loyalty_points INT DEFAULT 0,

    CONSTRAINT chk_loyalty_points
        CHECK (loyalty_points >= 0),

    CONSTRAINT fk_customers_user
        FOREIGN KEY (user_id) REFERENCES users(id)
);
```

```
CREATE TABLE categories (
   id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(100) NOT NULL,
   slug VARCHAR(150) NOT NULL UNIQUE,
   image VARCHAR(255),
   description TEXT,
   status VARCHAR(50) NOT NULL
);

CREATE TABLE products (
   id INT PRIMARY KEY AUTO_INCREMENT,
   seller_id INT NOT NULL,
   category_id INT NOT NULL,
   name VARCHAR(150) NOT NULL,
   slug VARCHAR(200) NOT NULL UNIQUE,
   description TEXT,
   price DECIMAL(10,2) NOT NULL,
   discount_price DECIMAL(10,2),
   thumbnail VARCHAR(255),
   status VARCHAR(50) NOT NULL,
```

```
    CONSTRAINT chk_product_price
        CHECK (price > 0),

    CONSTRAINT chk_discount_price
        CHECK (discount_price >= 0),

    CONSTRAINT fk_products_seller
        FOREIGN KEY (seller_id) REFERENCES sellers(id),

    CONSTRAINT fk_products_category
        FOREIGN KEY (category_id) REFERENCES categories(id)
);

CREATE TABLE product_variants (
    id INT PRIMARY KEY AUTO_INCREMENT,
    product_id INT NOT NULL,
    size VARCHAR(50),
    color VARCHAR(50),
    sku VARCHAR(100) NOT NULL UNIQUE,
    price DECIMAL(10,2) NOT NULL,
    stock INT NOT NULL,
```

```sql
    CONSTRAINT chk_variant_price
        CHECK (price > 0),

    CONSTRAINT chk_variant_stock
        CHECK (stock >= 0),

    CONSTRAINT fk_product_variants_product
        FOREIGN KEY (product_id) REFERENCES products(id)
);

CREATE TABLE inventory (
    id INT PRIMARY KEY AUTO_INCREMENT,
    product_id INT NOT NULL,
    stock INT NOT NULL,
    reserved_stock INT DEFAULT 0,
    warehouse_location VARCHAR(255),
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,

    CONSTRAINT chk_inventory_stock
        CHECK (stock >= 0),
```

```
    CONSTRAINT chk_reserved_stock
        CHECK (reserved_stock >= 0),

    CONSTRAINT fk_inventory_product
        FOREIGN KEY (product_id) REFERENCES products(id)
);

CREATE TABLE product_images (
    id INT PRIMARY KEY AUTO_INCREMENT,
    product_id INT NOT NULL,
    image_url VARCHAR(255) NOT NULL,

    CONSTRAINT fk_product_images_product
        FOREIGN KEY (product_id) REFERENCES products(id)
);

CREATE TABLE addresses (
    id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT NOT NULL,
    full_name VARCHAR(100) NOT NULL,
    phone VARCHAR(20) NOT NULL,
    country_id INT NOT NULL,
```

```sql
    province_id INT NOT NULL,
    city_id INT NOT NULL,
    postal_code VARCHAR(20),
    address_line_1 VARCHAR(255) NOT NULL,
    address_line_2 VARCHAR(255),
    landmark VARCHAR(255),
    address_type VARCHAR(50) NOT NULL,

    CONSTRAINT fk_addresses_user
        FOREIGN KEY (user_id) REFERENCES users(id)
);

CREATE TABLE carts (
    id INT PRIMARY KEY AUTO_INCREMENT,
    customer_id INT NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT fk_carts_customer
        FOREIGN KEY (customer_id) REFERENCES customers(id)
);

CREATE TABLE cart_items (
```

```sql
    id INT PRIMARY KEY AUTO_INCREMENT,
    cart_id INT NOT NULL,
    product_id INT NOT NULL,
    quantity INT NOT NULL,

    CONSTRAINT chk_cart_quantity
        CHECK (quantity > 0),

    CONSTRAINT fk_cart_items_cart
        FOREIGN KEY (cart_id) REFERENCES carts(id),

    CONSTRAINT fk_cart_items_product
        FOREIGN KEY (product_id) REFERENCES products(id)
);

CREATE TABLE wishlists (
    id INT PRIMARY KEY AUTO_INCREMENT,
    customer_id INT NOT NULL,

    CONSTRAINT fk_wishlists_customer
        FOREIGN KEY (customer_id) REFERENCES customers(id)
);
```

```
CREATE TABLE wishlist_items (
    id INT PRIMARY KEY AUTO_INCREMENT,
    wishlist_id INT NOT NULL,
    product_id INT NOT NULL,

    CONSTRAINT fk_wishlist_items_wishlist
        FOREIGN KEY (wishlist_id) REFERENCES wishlists(id),

    CONSTRAINT fk_wishlist_items_product
        FOREIGN KEY (product_id) REFERENCES products(id)
);

CREATE TABLE orders (
    id INT PRIMARY KEY AUTO_INCREMENT,
    customer_id INT NOT NULL,
    address_id INT NOT NULL,
    order_number VARCHAR(100) NOT NULL UNIQUE,
    subtotal DECIMAL(10,2) NOT NULL,
    shipping_fee DECIMAL(10,2) DEFAULT 0,
    tax DECIMAL(10,2) DEFAULT 0,
    discount DECIMAL(10,2) DEFAULT 0,
```

```sql
    total_amount DECIMAL(10,2) NOT NULL,
    payment_method_id INT NOT NULL,
    payment_status_id INT NOT NULL,
    order_status_id INT NOT NULL,

    CONSTRAINT chk_order_total
        CHECK (total_amount >= 0),

    CONSTRAINT fk_orders_customer
        FOREIGN KEY (customer_id) REFERENCES customers(id),

    CONSTRAINT fk_orders_address
        FOREIGN KEY (address_id) REFERENCES addresses(id)
);

CREATE TABLE order_items (
    id INT PRIMARY KEY AUTO_INCREMENT,
    order_id INT NOT NULL,
    product_id INT NOT NULL,
    seller_id INT NOT NULL,
    quantity INT NOT NULL,
    price DECIMAL(10,2) NOT NULL,
```

```sql
    subtotal DECIMAL(10,2) NOT NULL,

    CONSTRAINT chk_order_quantity
        CHECK (quantity > 0),

    CONSTRAINT chk_order_price
        CHECK (price > 0),

    CONSTRAINT fk_order_items_order
        FOREIGN KEY (order_id) REFERENCES orders(id),

    CONSTRAINT fk_order_items_product
        FOREIGN KEY (product_id) REFERENCES products(id),

    CONSTRAINT fk_order_items_seller
        FOREIGN KEY (seller_id) REFERENCES sellers(id)
);

CREATE TABLE payments (
    id INT PRIMARY KEY AUTO_INCREMENT,
    order_id INT NOT NULL,
    transaction_id VARCHAR(150) NOT NULL UNIQUE,
```

```sql
    payment_method_id INT NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    payment_status_id INT NOT NULL,

    CONSTRAINT chk_payment_amount
        CHECK (amount > 0),

    CONSTRAINT fk_payments_order
        FOREIGN KEY (order_id) REFERENCES orders(id)
);

CREATE TABLE reviews (
    id INT PRIMARY KEY AUTO_INCREMENT,
    customer_id INT NOT NULL,
    product_id INT NOT NULL,
    rating INT NOT NULL,
    comment TEXT,

    CONSTRAINT chk_review_rating
        CHECK (rating >= 1 AND rating <= 5),

    CONSTRAINT uq_customer_product_review
```

```sql
        UNIQUE(customer_id, product_id),

    CONSTRAINT fk_reviews_customer
        FOREIGN KEY (customer_id) REFERENCES customers(id),

    CONSTRAINT fk_reviews_product
        FOREIGN KEY (product_id) REFERENCES products(id)
);

CREATE TABLE notifications (
    id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT NOT NULL,
    title VARCHAR(255) NOT NULL,
    message TEXT NOT NULL,
    type VARCHAR(50) NOT NULL,
    is_read BOOLEAN DEFAULT FALSE,

    CONSTRAINT fk_notifications_user
        FOREIGN KEY (user_id) REFERENCES users(id)
);
```

Data Population (DML)

Enable Local Infile

```
SET GLOBAL local_infile = 1;
```

UPDATE Statement with WHERE Condition

```
UPDATE products
SET price = price + 100
WHERE category_id = 1;
```

DELETE Statement with WHERE Condition

```
DELETE FROM reviews
WHERE rating = 1;
```

COUNT(*) Validation Queries

```
SELECT COUNT(*) AS total_roles FROM roles;
SELECT COUNT(*) AS total_user_status FROM user_status;
SELECT COUNT(*) AS total_users FROM users;
SELECT COUNT(*) AS total_sellers FROM sellers;
SELECT COUNT(*) AS total_customers FROM customers;
SELECT COUNT(*) AS total_categories FROM categories;
SELECT COUNT(*) AS total_products FROM products;
SELECT COUNT(*) AS total_orders FROM orders;
SELECT COUNT(*) AS total_order_items FROM order_items;
SELECT COUNT(*) AS total_reviews FROM reviews;
```

NULL Check Queries

```
SELECT * FROM users
WHERE name IS NULL
   OR email IS NULL
   OR role_id IS NULL;

SELECT * FROM products
WHERE seller_id IS NULL
```

```
  OR category_id IS NULL
  OR price IS NULL;

SELECT * FROM orders
WHERE customer_id IS NULL
  OR order_number IS NULL;
```

JOIN-Based Foreign Key Integrity Checks

```
SELECT p.id AS product_id,
     p.name AS product_name,
     s.shop_name
FROM products p
JOIN sellers s
ON p.seller_id = s.id;

SELECT o.order_number,
     c.id AS customer_id,
     u.name AS customer_name
FROM orders o
JOIN customers c
ON o.customer_id = c.id
JOIN users u
ON c.user_id = u.id;

SELECT oi.id AS order_item_id,
     p.name AS product_name,
     oi.quantity,
     oi.subtotal
FROM order_items oi
JOIN products p
ON oi.product_id = p.id;
```